

Problem Set 4

POLS W4732

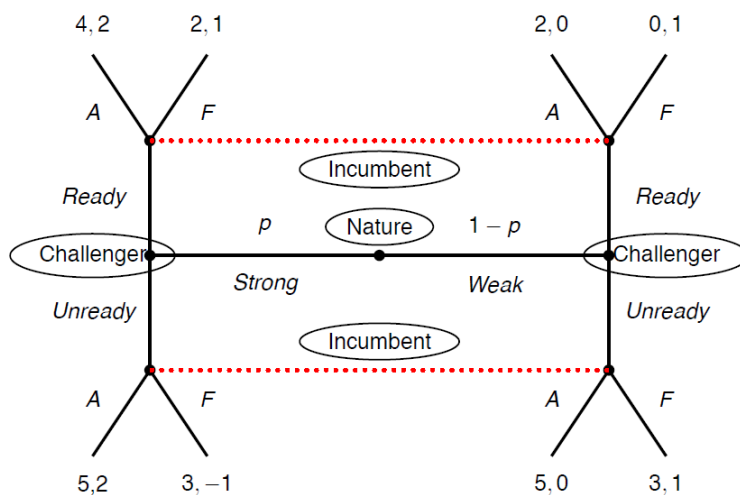
Fall 2026

Due November 6 before the TA section

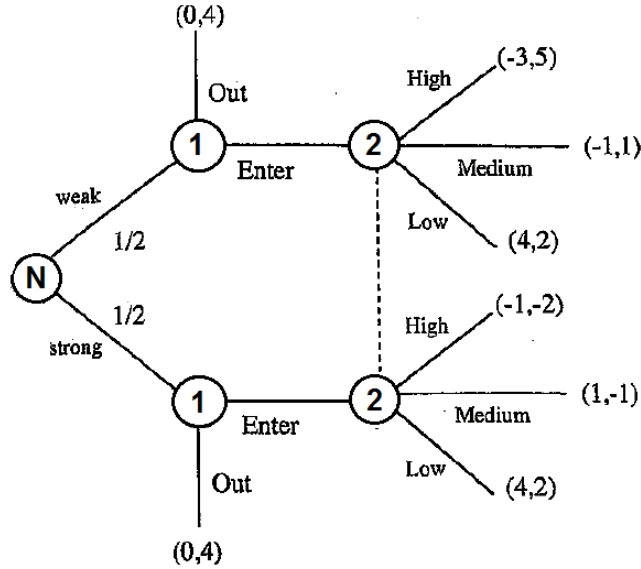
Instructions. For each formal claim, include two or three algebraic inequalities or best-response comparisons that establish it. Unless directly following notation from slides or the text of the problem, define your notation.

Artificial-intelligence policy. Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

Exercise 1. Consider the crisis bargaining game:



Show that if one modifies one payoff to one player in one terminal node of the crisis bargaining game, the separating equilibrium no longer exists



Exercise 2. Consider the following game with an informed entrant, where $a_1(W)$ denotes the pure strategy of a Weak type of Pl. 1, $a_1(S)$ the pure strategy of a Strong type Pl. 1. Pl. 2 only observes an action a_1 and chooses her best response $a_2(a_1)$.

1. Describe the set of terminal histories and the set of non-terminal histories of this game
2. For each player, describe the information sets of this game
3. Does the profile $(a_1(S) = a_1(W) = \text{Out}, a_2 = \text{High})$ constitute a Perfect Bayesian Equilibrium?
4. Does the profile $(a_1(S) = a_1(W) = \text{Enter}, a_2 = \text{Low})$ constitute a Perfect Bayesian Equilibrium?
5. Does the profile $(a_1(S) = \text{Enter}, a_1(W) = \text{Out}, a_2 = \text{Low})$ constitute a Perfect Bayesian Equilibrium?

Exercise 3. There are two types of worker. Type θ has a marginal value product of $v_\theta = \theta$, where $\theta \in \{1, 2\}$. The cost of sending signal z for the low type is $C_1(z) = z$ and for the high type is $C_2(z) = z(1 - \epsilon)$. With probability π , the worker is a low type.

The equilibrium wage is going to be the expected marginal value product of the worker.¹

1. Show that there is a continuum of PBEs in which Type 1 chooses $z = 0$ and Type 2 chooses signal $z \in Z \equiv (1, 1/(1 - \epsilon))$
2. Consider now the possibility of a pooling equilibrium. What are the possible values of z compatible with a pooling equilibrium?

¹One way of justifying this is to assume that there are two firms competing à la Bertrand for the worker.

3. *What is the highest equilibrium payoff to each type of worker from a pooling equilibrium?
Can we conclude that Type 2 is always better off under a separating equilibrium?*